Risk	Initial L×I	Mitigation / response	Residual L×I
R1 – No suitable labelled dataset, or unclear provenance or licence	3×4 = 12	Assess more than one candidate dataset early; verify labels, source, licence and terms before committing. Confirm what “phishing” means in each dataset, as spam and phishing are not the same.	2×3 = 6
R2 – Project scope becomes too large	4×5 = 20	Fix the scope at three models, one main dataset, one primary explanation approach (LIME plus LR coefficients) and a core prototype. Treat SHAP, extra datasets and extra features as stretch goals. Agree scope with the supervisor and review it each sprint.	2×4 = 8
R3 – Tasks fall behind schedule	4×4 = 16	Weekly tracking against the Gantt chart; supervisor review each sprint; identify dependencies early; re-plan when slippage occurs; use the Christmas buffer.	2×4 = 8
R4 – Insufficient time for dissertation	3×5 = 15	Write notes and methods during development; keep a dissertation outline; reserve dedicated writing and proofreading time (M9, M11).	2×4 = 8
R5 – Difficulty understanding or implementing libraries	3×3 = 9	Build small examples first; use official scikit-learn, LIME and Flask documentation; ask the supervisor early.	2×2 = 4
R6 – Model performance is poor	3×4 = 12	Check labels and preprocessing; establish a baseline; tune fairly with cross-validation; inspect errors; report limitations rather than over-claiming.	2×4 = 8
R7 – Data leakage or duplicates inflate results	3×5 = 15	Remove duplicates; split before fitting any transformation; use scikit-learn pipelines; keep the test set untouched until final evaluation.	2×4 = 8
R8 – Explanations are misleading or hard to implement	3×4 = 12	Start with LR coefficients (simple and transparent); prototype LIME early on one model; compare explanations with error cases; state method limitations; SHAP only if suitable.	2×3 = 6
R9 – Flask integration or demo fails	3×4 = 12	Integrate early; test the full workflow regularly; keep a stable tagged release; prepare sample inputs and setup instructions; record a backup demo video.	2×3 = 6
R10 – Privacy or licensing issue	2×4 = 8	Use permitted research data; check original terms; avoid real personal emails; do not store user input by default; acknowledge sources.	1×4 = 4
R11 – Personal circumstances or illness interrupt work	2×4 = 8	Realistic schedule with buffer; frequent backups; tell the supervisor early; use mitigating circumstances procedures if needed.	2×3 = 6
R12 – Loss or corruption of code, data or report files	2×4 = 8	Commit regularly to GitHub; keep a separate secure backup of code, data and documents.	1×4 = 4
R13 – Supervisor feedback arrives later than expected	3×3 = 9	Send concise questions and draft sections in advance; continue low-risk tasks while waiting; record decisions and revise the plan.	2×3 = 6
R14 – Prototype gives users false reassurance	3×4 = 12	Show a clear limitation notice; avoid “safe” wording; advise normal security checks; report false-negative rates honestly.	2×3 = 6
R15 – Random Forest or SVM training is slow or hard to tune on high-dimensional TF-IDF features	3×3 = 9	Limit vocabulary size (for example max_features); start with a data subset; use sensible default hyperparameters before tuning; drop extensive tuning if time is short.	2×2 = 4